Literature Survey

Overview

The literature survey focuses on studying existing crop recommendation systems, smart farming technologies, and machine learning applications in agriculture. Research papers, journals, and online resources were reviewed to understand how artificial intelligence can improve crop selection and farming efficiency.

Key Findings

The study explored various machine learning algorithms, including **Decision Tree**, **Random Forest**, **K-Nearest Neighbors (KNN)**, **Logistic Regression**, and **K-Means Clustering**. These algorithms are widely used for crop recommendation, soil analysis, and yield prediction based on environmental and soil conditions.

The survey also examined data preprocessing techniques, feature selection, and model evaluation methods to improve prediction accuracy. Existing agricultural systems highlighted the importance of using data-driven approaches for sustainable farming while identifying limitations such as limited datasets, lower prediction accuracy, and scalability challenges.

Conclusion

The literature survey provided valuable insights into current agricultural technologies and machine learning techniques. These findings helped in selecting suitable methodologies and designing the **OptiCrop** system to deliver accurate crop recommendations and support efficient, sustainable farming practices.